

Alex McIntosh

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SUMMARY

Operationalizing **interpretable machine learning**, **explainable AI**, and **observable agentic networks** reliably at scale.

EXPERIENCE

Pioneer Management Consulting, Consultant, Artificial Intelligence Minneapolis, MN
Designing custom AI architectures and developing conversational analytics to drive sustainable enterprise AI adoption. June 2026 – present
4 months

- Design AI architectures delivering scalable predictive analytics and anomaly detection for enterprise clients.
- Develop AI-powered conversational analytics and generative domain bots.
- Implement model explainability into technical designs to ensure client trust and mitigate risk.

Optum, AI/ML Engineer Associate Eden Prairie, MN
Architected and built enterprise-scale agentic voice AI systems with end-to-end observability, delivering AI tools to 400K+ employees. Aug 2025 – June 2026
11 months

- Shipped an agentic voice AI frontend using LiveKit to expose a conversational frontend to a multi-agent system
- Automated performance evaluation for LLM agents and RAG pipelines
- Shipped a multi-agent system using A2A, MCP, and ADK to automate admin tasks

Optum, Data Engineer Associate Eden Prairie, MN
Architected and deployed enterprise-scale agentic AI systems with end-to-end observability, enhancing productivity for over 400K employees Aug 2024 – Aug 2025
1 year 1 month

- Automated LLM evaluation with scoring pipelines to monitor consistency, accuracy, safety, latency, and fairness
- Refactored Fabric Spark ETL into reusable patterns, cutting failure rates from 12-15% to <1%
- Mentored an intern team that won first place in the Optum-wide Shark Tank competition with an agentic AI solution

Optum, Data Science and Engineering Associate Eden Prairie, MN
Developed and operationalized ML models and multi-agent systems to drive analytics and improve enterprise decision-making. July 2023 – July 2024
1 year 1 month

- Trained a state-of-the-art time-series forecasting model (Prophet + Random Forest) to detect operational anomalies with >90% precision.
- Orchestrated multi-agent AI systems to perform statistical analysis dashboard generation
- Delivered a conversational AI self-service analytics solution

EDUCATION

Master of Science **Georgia Institute of Technology**, Computer Science July 2027
Machine Learning specialization

MicroMasters **MITx**, Statistics and Data Science Dec 2023

Bachelor of Arts **Saint John's University**, Physics May 2021
Minored in Music & Mathematics • Phi Beta Kappa • Men's Chorus President

PRODUCTION INFRASTRUCTURE MLOPS

Production Infrastructure & MLOps: Docker, OpenTelemetry, FastAPI, GitHub Actions (CI/CD), Microsoft Azure, Google Cloud Platform, Git

SKILLS

Datascience and ML: Python, SQL, R, PyTorch, TensorFlow, Scikit-learn, PySpark, MLFlow, Statistics and Probability

Agentic AI & RAG: Model Context Protocol (MCP), Agent Development Kit (ADK), Agent-to-Agent (A2A) Protocol, Semantic Kernel, Retrieval-Augmented Generation (RAG), LiveKit

Data & Cloud: Microsoft Fabric, Databricks, Delta Lake, AirFlow, ChromaDB, Git, Docker, Microsoft Azure, Google Cloud Platform, OpenTelemetry